

Gaussian Process Regression & Classification: Algorithms 2.1, 3.1, and 3.2

From *Gaussian Processes for Machine Learning* — Rasmussen & Williams

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Abstract

We implement Algorithms 2.1 (GP Regression), 3.1 (Laplace Approximation mode-finding via Newton’s method), and 3.2 (Predictions for binary Laplace GP Classification) from scratch using only NumPy and SciPy for linear algebra. The GP regressor is evaluated on a 1-D toy function and the 5-D Friedman #1 dataset, achieving $R^2 = 0.9206$ on the latter. For classification, the Laplace-approximated GP classifier is tested on Two Moons (98.33% accuracy) and Iris binary (97.78% accuracy). We visualise predictive covariance in 2-D, compare MAP versus averaged predictions, and conduct hyperparameter ablations over the kernel length scale ℓ , noise level σ_n , and kernel type. The averaged prediction $\bar{\pi}_*$ (Eq. 3.25) is shown to be more conservative than the MAP prediction $\hat{\pi}_* = \sigma(\hat{f}_*)$ in regions of high latent variance, providing better-calibrated uncertainty estimates.

1 Introduction

A Gaussian Process (GP) defines a distribution over functions: any finite collection of function values is jointly Gaussian. GPs are attractive because they provide a full predictive distribution (mean and variance) rather than just a point estimate, and they allow principled model selection via the log marginal likelihood.

We implement three algorithms from Rasmussen & Williams:

- **Algorithm 2.1:** GP regression using Cholesky-based inference. Given training inputs X , targets $\mathbf{y}$, a covariance function k , and noise variance σ_n^2 , it returns the predictive mean, variance, and log marginal likelihood.
- **Algorithm 3.1:** Mode-finding for the Laplace approximation in binary GP classification. Since the logistic likelihood is non-Gaussian, the posterior $p(\mathbf{f}|X, \mathbf{y})$ is intractable; the Laplace method approximates it with a Gaussian centred at the posterior mode $\hat{\mathbf{f}}$, found via Newton iterations.
- **Algorithm 3.2:** Predictions for binary Laplace GP classification. Given the mode from Algorithm 3.1, it computes the latent mean $\hat{f}_*$, latent variance $\mathbb{V}[f_*]$, and the averaged predictive class probability $\bar{\pi}_*$.

2 Algorithm Descriptions

2.1 Algorithm 2.1: GP Regression

Inputs: X (training inputs), $\mathbf{y}$ (targets), k (covariance function), σ_n^2 (noise level), $\mathbf{x}_*$ (test input).

1. $L := \text{cholesky}(K + \sigma_n^2 I)$
2. $\alpha := L^\top \backslash (L \backslash \mathbf{y})$
3. $\bar{f}_* := \mathbf{k}_*^\top \alpha$ (predictive mean, Eq. 2.25)
4. $\mathbf{v} := L \backslash \mathbf{k}_*$
5. $\mathbb{V}[f_*] := k(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{v}^\top \mathbf{v}$ (predictive variance, Eq. 2.26)
6. $\log p(\mathbf{y}|X) := -\frac{1}{2} \mathbf{y}^\top \alpha - \sum_i \log L_{ii} - \frac{n}{2} \log 2\pi$ (log marginal likelihood, Eq. 2.30)

The Cholesky decomposition ensures numerical stability and costs $O(n^3)$. No iterative optimisation or gradient descent is involved; inference is a single closed-form computation.

2.2 Algorithm 3.1: Laplace Approximation (Newton's Method)

For classification with logistic likelihood $p(y_i|f_i) = \sigma(y_i f_i)$, the posterior is non-Gaussian. We approximate it as $q(\mathbf{f}|X, \mathbf{y}) = \mathcal{N}(\hat{\mathbf{f}}, A^{-1})$ where $\hat{\mathbf{f}}$ is the posterior mode and $A = -\nabla \nabla \log p(\mathbf{f}|X, \mathbf{y})|_{\mathbf{f}=\hat{\mathbf{f}}}$.

Inputs: K (covariance matrix), $\mathbf{y}$ (± 1 targets), $p(\mathbf{y}|\mathbf{f})$ (likelihood).

1. Initialise $\mathbf{f} := \mathbf{0}$.
2. **Repeat** (Newton iteration):
 - (a) $W := -\nabla \nabla \log p(\mathbf{y}|\mathbf{f})$ (diagonal, $W_{ii} = \pi_i(1 - \pi_i)$)
 - (b) $L := \text{cholesky}(I + W^{1/2} K W^{1/2})$, $B = I + W^{1/2} K W^{1/2}$
 - (c) $\mathbf{b} := W \mathbf{f} + \nabla \log p(\mathbf{y}|\mathbf{f})$
 - (d) $\mathbf{a} := \mathbf{b} - W^{1/2} L^\top \backslash (L \backslash (W^{1/2} K \mathbf{b}))$ (Eq. 3.18 via 3.27)
 - (e) $\mathbf{f} := K \mathbf{a}$
3. **Until** convergence.
4. $\log q(\mathbf{y}|X, \theta) := -\frac{1}{2} \mathbf{a}^\top \mathbf{f} + \log p(\mathbf{y}|\mathbf{f}) - \sum_i \log L_{ii}$ (Eq. 3.32)

The logistic likelihood gives: $\nabla_i \log p(y_i|f_i) = t_i - \pi_i$ where $t_i = (y_i + 1)/2$ and $\pi_i = \sigma(f_i)$, and $W_{ii} = \pi_i(1 - \pi_i)$. The matrix $B = I + W^{1/2} K W^{1/2}$ is symmetric positive definite, making Cholesky decomposition numerically safe.

2.3 Algorithm 3.2: Predictions for Binary Laplace GP Classification

Inputs: $\hat{\mathbf{f}}$ (mode from Alg. 3.1), X , $\mathbf{y}$, k , $p(\mathbf{y}|\mathbf{f})$, $\mathbf{x}_*$ (test input).

1. $W := -\nabla \nabla \log p(\mathbf{y}|\hat{\mathbf{f}})$, $L := \text{cholesky}(I + W^{1/2} K W^{1/2})$
2. $\bar{f}_* := \mathbf{k}(\mathbf{x}_*)^\top \nabla \log p(\mathbf{y}|\hat{\mathbf{f}})$ (latent mean, Eq. 3.21)
3. $\mathbf{v} := L \backslash (W^{1/2} \mathbf{k}(\mathbf{x}_*))$
4. $\mathbb{V}[f_*] := k(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{v}^\top \mathbf{v}$ (latent variance, Eq. 3.24 via 3.29)
5. $\bar{\pi}_* := \int \sigma(z) \mathcal{N}(z|\bar{f}_*, \mathbb{V}[f_*]) dz$ (averaged prediction, Eq. 3.25)

The integral in step 5 is approximated using the probit approximation: $\bar{\pi}_* \approx \sigma(\bar{f}_* / \sqrt{1 + \pi \mathbb{V}[f_*]/8})$.

3 Datasets

Table 1: Datasets used in experiments.

Task	Dataset	Train	Test	Features	Notes
Regression	1-D Toy	20	300	1	$f(x) = \sin(x) + 0.5 \cos(2x)$
Regression	Friedman #1	300	100	5	Synthetic, known ground truth
Regression	2-D Toy	30	40×40 grid	2	$f = \sin(x_1) \cos(x_2)$
Classification	Two Moons	140	60	2	Nonlinear boundary
Classification	Iris (binary)	105	45	2	Setosa vs. rest

1-D Toy: Demonstrates GP regression posterior with confidence bands. 20 noisy observations from $f(x) = \sin(x) + 0.5 \cos(2x)$ with $\sigma_n = 0.3$.

Friedman #1: The true function is $y = 10 \sin(\pi x_1 x_2) + 20(x_3 - 0.5)^2 + 10x_4 + 5x_5 + \varepsilon$. Tests the GP in higher dimensions with a known regression function.

2-D Toy: 30 points in $\mathbb{R}^2$ from $f = \sin(x_1) \cos(x_2)$. Used specifically for predictive covariance visualisation.

Two Moons: 200 points from the two-moons distribution with Gaussian noise ($\sigma = 0.2$), standardised. A nonlinear binary classification benchmark.

Iris (binary): The first two features of the Iris dataset, with Setosa as class +1 and the rest as class -1. Standardised.

4 Results

4.1 GP Regression (Algorithm 2.1)

Table 2: GP Regression results (RBF kernel).

Dataset	MSE	R^2	Log ML	Kernel
1-D Toy	–	–	-18.36	RBF($\ell = 1.0, \sigma_f = 1.0$)
Friedman #1 (5-D)	1.9099	0.9206	-577.79	RBF($\ell = 2.0, \sigma_f = 5.0$)

On the 1-D toy data, the GP posterior mean closely tracks the true function, with the $\pm 2\sigma$ confidence band widening away from the training data and narrowing near observations (Figure 1). The log marginal likelihood is -18.36.

On Friedman #1, the GP achieves $R^2 = 0.9206$ and $\text{MSE} = 1.91$ despite using only 300 training points in 5 dimensions. The uncertainty calibration plot (Figure 3) shows that the empirical coverage tracks the ideal Gaussian line reasonably well, indicating that the predictive variance is meaningful.

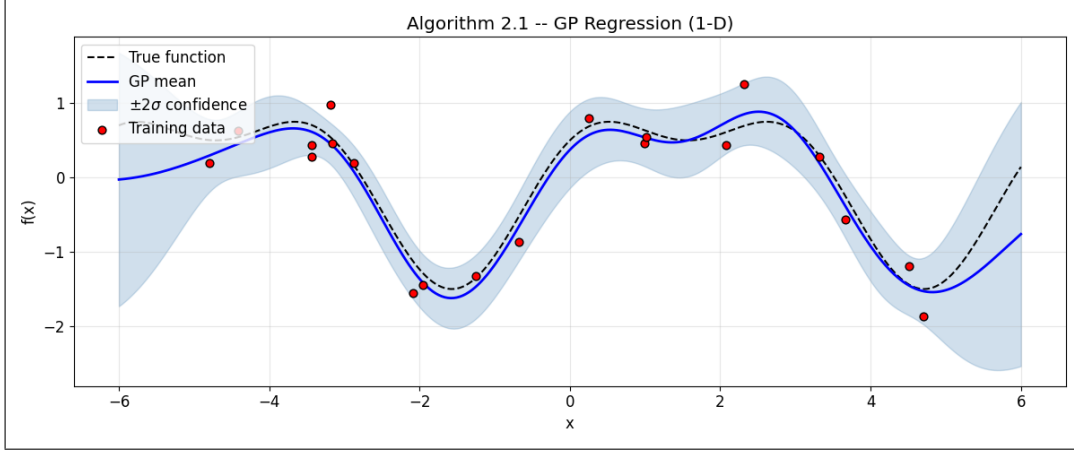


Figure 1: Algorithm 2.1: GP Regression on 1-D toy data. The GP mean (blue) closely follows the true function (dashed), with the $\pm 2\sigma$ confidence band widening away from training points (red dots).

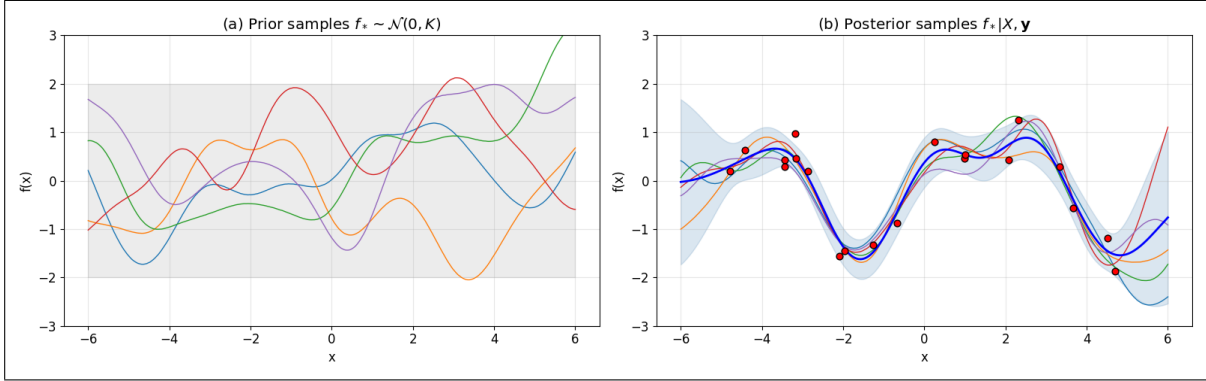


Figure 2: (a) Five functions sampled from the GP prior $\mathcal{N}(\mathbf{0}, K)$. (b) Five posterior samples conditioned on the 20 training points, with the posterior mean and $\pm 2\sigma$ band.

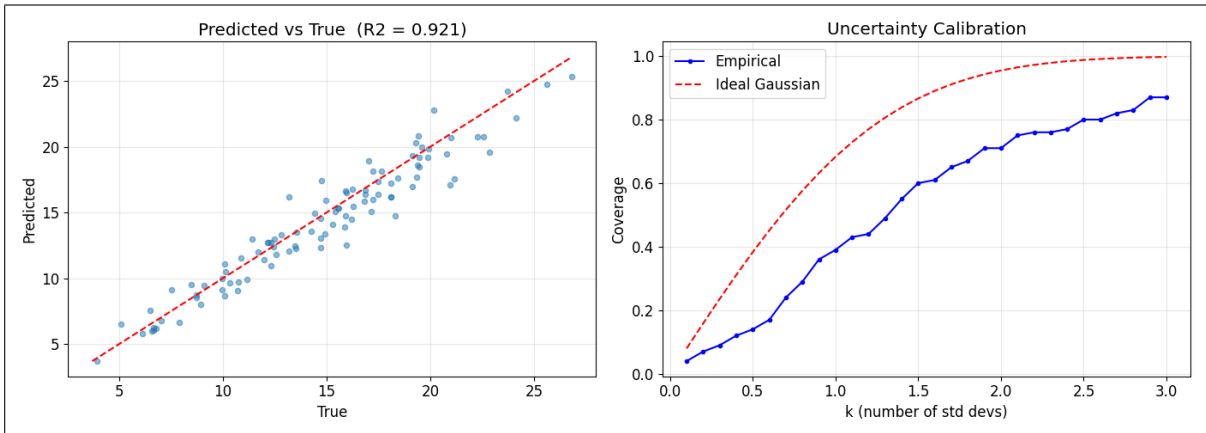


Figure 3: Friedman #1 (5-D). Left: predicted vs. true values ($R^2 = 0.921$). Right: uncertainty calibration showing empirical coverage against the ideal Gaussian.

4.2 2-D Predictive Covariance

On the 2-D toy dataset, we computed the full posterior covariance matrix at a 40×40 grid of test locations. Figure 4 shows three views: (a) the predictive mean, (b) the predictive variance $\mathbb{V}[f_*]$, and (c) the covariance of a test point $\mathbf{x}_* = (0.10, 0.10)$ with all other grid locations. At this test point, the predictive mean is $\bar{f}_* = -0.0206$ and the predictive variance is $\mathbb{V}[f_*] = 0.0105$.

The variance is lowest near training points and highest in data-sparse regions (corners of the grid). The covariance slice shows that nearby points are positively correlated, with correlation decaying smoothly with distance, consistent with the RBF kernel’s smoothness properties.

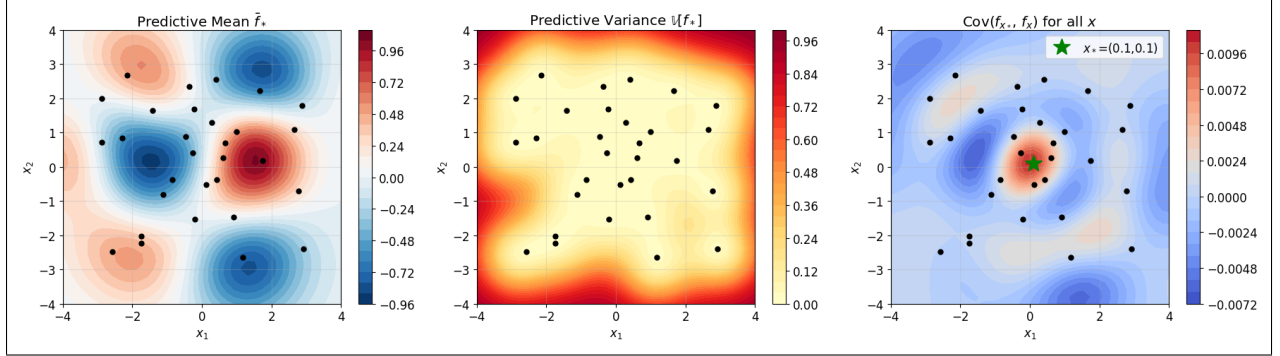


Figure 4: 2-D toy dataset. (a) Predictive mean $\bar{f}_*$. (b) Predictive variance $\mathbb{V}[f_*]$. (c) Covariance between a test point $\mathbf{x}_* = (0.1, 0.1)$ (green star) and all other locations.

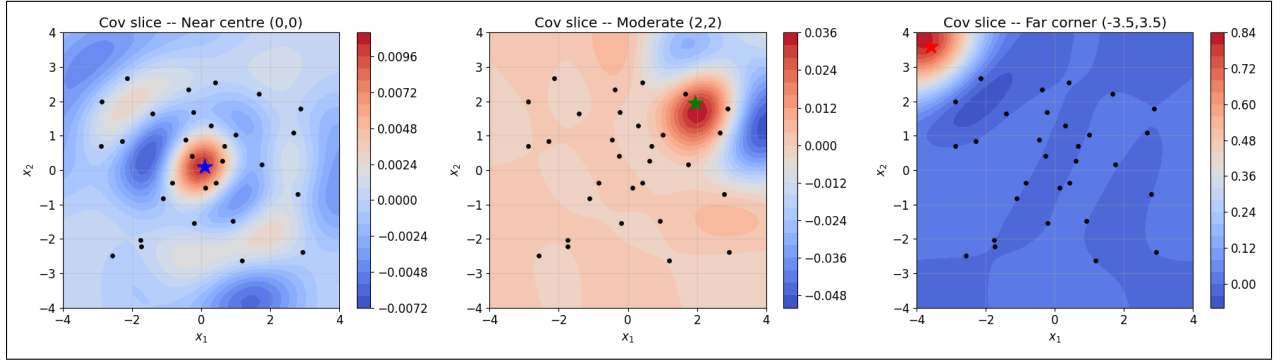


Figure 5: Covariance slices for three test points at different distances from training data. Near the centre, the covariance is concentrated. Far from data (corner), the covariance spread is broader and the magnitude is larger, reflecting higher uncertainty.

4.3 Newton Convergence (Algorithm 3.1)

Algorithm 3.1 converges rapidly: on a 50-point demo dataset, the Newton iteration reached the posterior mode in only 5 iterations (Figure 6). The Newton objective $-\frac{1}{2}\mathbf{a}^\top \mathbf{f} + \log p(\mathbf{y}|\mathbf{f})$ increases monotonically, jumping from ≈ -20.25 to ≈ -20.09 in the first two steps and saturating thereafter. The approximate log marginal likelihood (Eq. 3.32) at convergence is $\log q(\mathbf{y}|X, \theta) = -22.79$, which is lower than the objective because it includes the additional complexity penalty $-\sum_i \log L_{ii}$ from the determinant of $B = I + W^{1/2}KW^{1/2}$.

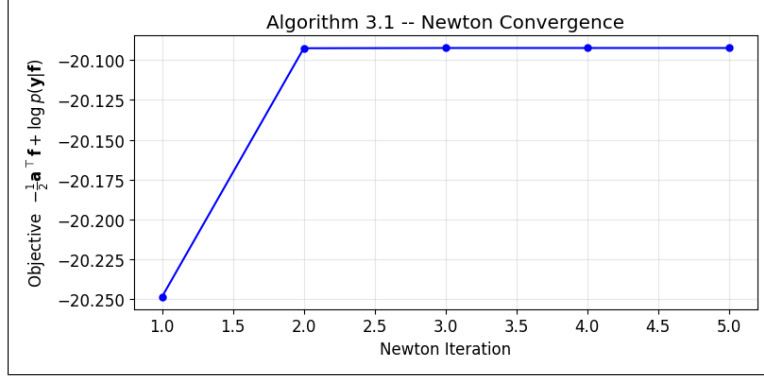


Figure 6: Algorithm 3.1: Newton convergence. The objective saturates within 3–4 iterations, reaching the posterior mode $\hat{\mathbf{f}}$ in 5 steps.

4.4 GP Classification (Algorithms 3.1 + 3.2)

Table 3: GP Classification results (RBF kernel, logistic likelihood).

Dataset	Accuracy	Macro F1	Newton Iters	Kernel
Two Moons	0.9833	0.98	5	$\text{RBF}(\ell = 0.8, \sigma_f = 1.0)$
Iris (binary)	0.9778	—	7	$\text{RBF}(\ell = 1.0, \sigma_f = 1.5)$

On Two Moons, the GP classifier achieves 98.33% test accuracy with precision 0.97/1.00 and recall 1.00/0.96 for the two classes respectively. On Iris (binary), accuracy is 97.78% with 7 Newton iterations.

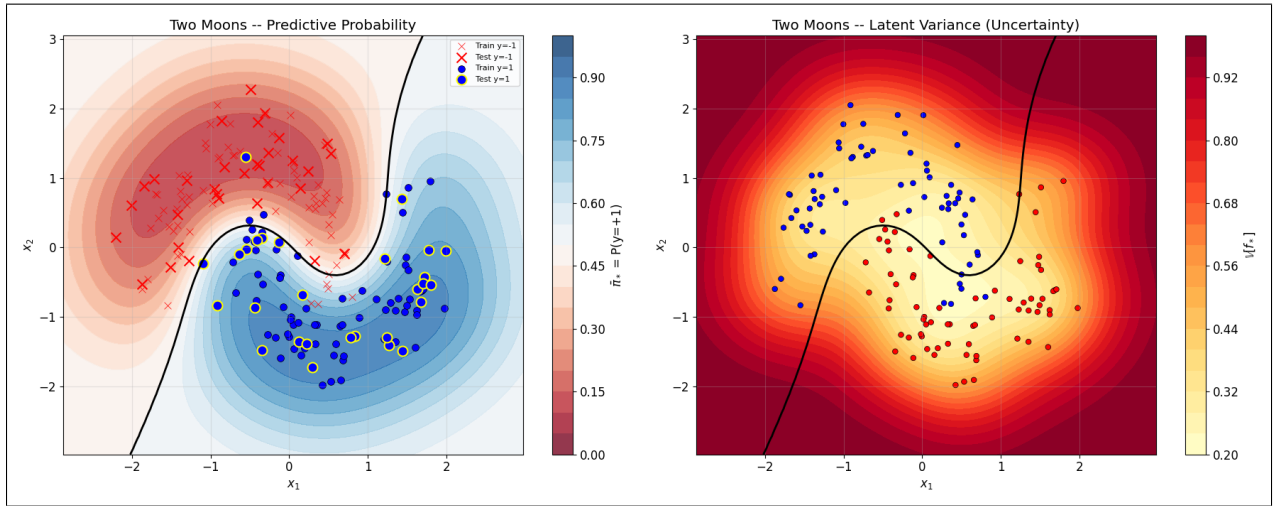


Figure 7: Two Moons. Left: predictive probability $\bar{\pi}_*$ with decision boundary (black line at 0.5). Right: latent variance $\mathbb{V}[f_*]$ showing high uncertainty near the boundary and far from data.

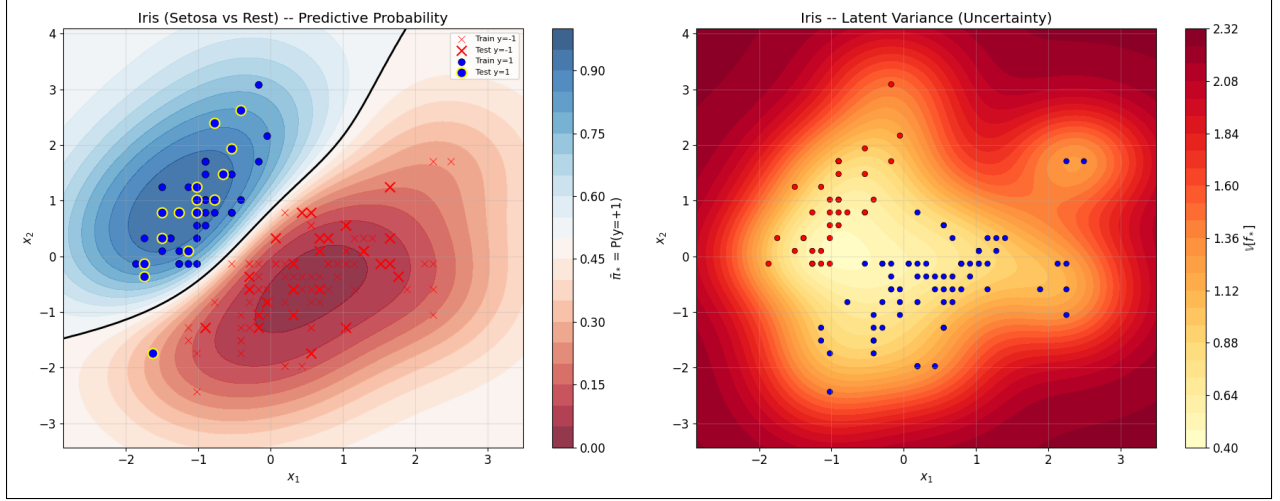


Figure 8: Iris (Setosa vs. rest). Left: predictive probability. Right: latent variance. The two classes are well-separated; uncertainty is concentrated along the boundary and at the edges.

4.5 MAP vs. Averaged Prediction

Figure 9 compares the MAP prediction $\hat{\pi}_* = \sigma(\bar{f}_*)$ with the averaged prediction $\bar{\pi}_* = \int \sigma(z) \mathcal{N}(z|\bar{f}_*, \mathbb{V}[f_*]) dz$ on Two Moons. The MAP prediction ignores latent variance and produces sharper probability boundaries, while the averaged prediction is more conservative (closer to 0.5) in uncertain regions. The difference $|\bar{\pi}_* - \hat{\pi}_*|$ is largest near the decision boundary and far from training data, precisely where $\mathbb{V}[f_*]$ is high.

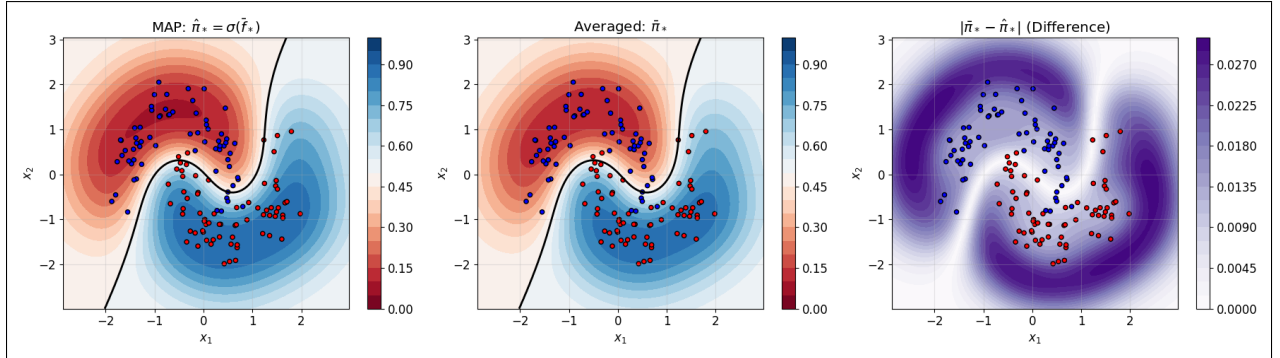


Figure 9: MAP vs. averaged prediction on Two Moons. Left: MAP $\hat{\pi}_* = \sigma(\bar{f}_*)$. Centre: averaged $\bar{\pi}_*$ (Eq. 3.25). Right: absolute difference. The averaged prediction is more conservative where uncertainty is high.

5 Hyperparameter Ablations

5.1 Effect of Length Scale ℓ on GP Regression

Figure 10 shows GP regression fits for $\ell \in \{0.2, 0.5, 1.0, 2.0, 5.0\}$. Small ℓ (0.2) produces a highly wiggly fit that overfits noise, while large ℓ (5.0) oversmooths and fails to capture the true function's variation.

The log marginal likelihood (LML) provides a principled criterion for selecting ℓ without cross-validation. Figure 11 shows LML as a function of ℓ ; the optimum occurs at $\ell \approx 0.95$, balancing data fit (first term of Eq. 2.30) against model complexity (second term).

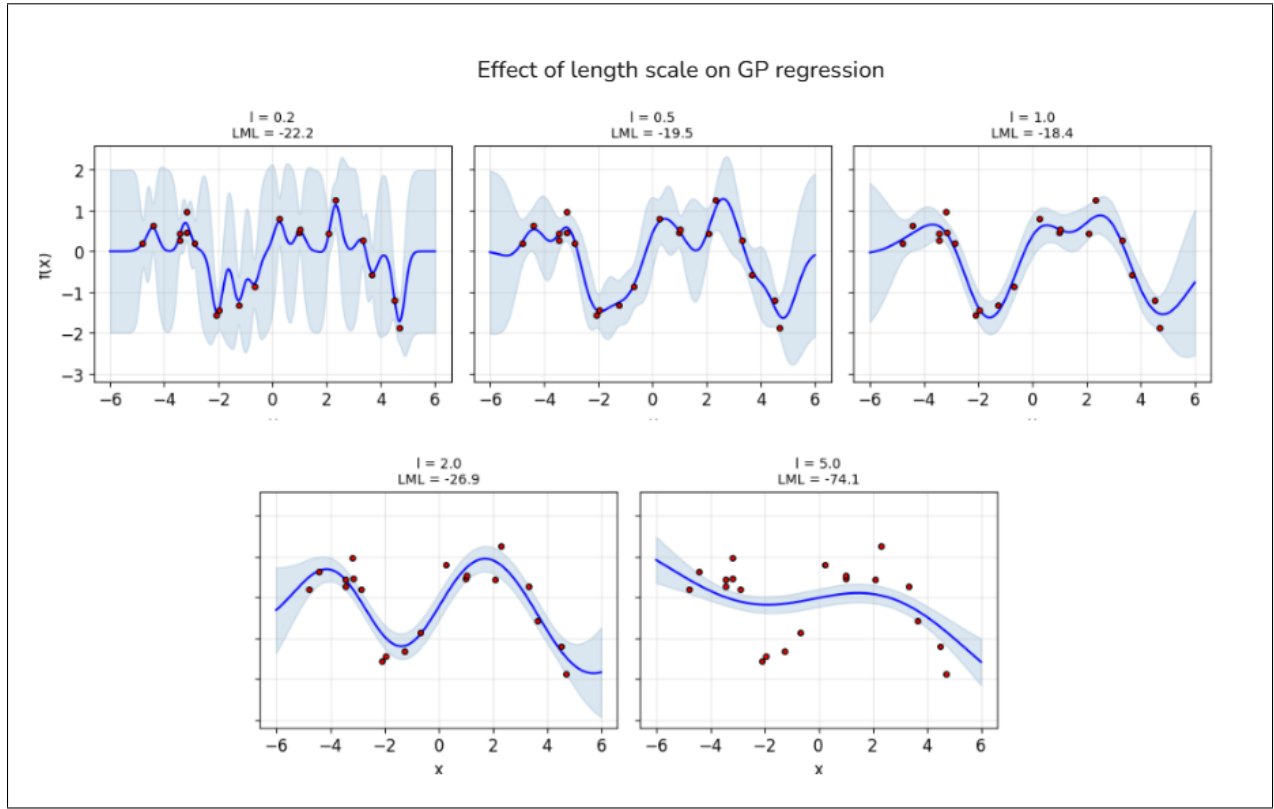


Figure 10: Effect of length scale ℓ on GP regression. Small ℓ overfits; large ℓ undersmooths. The log marginal likelihood (LML) values are shown in each panel title.

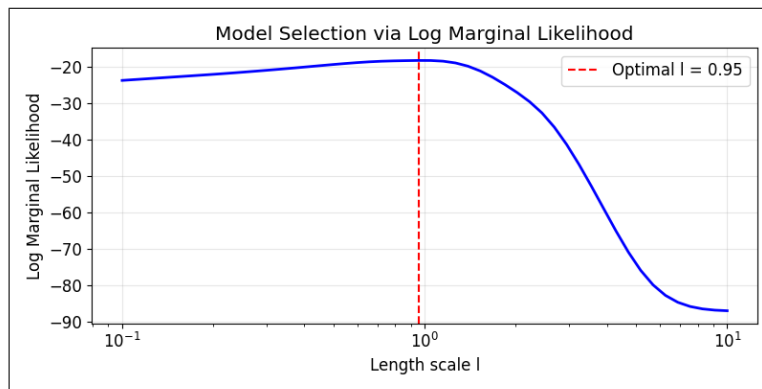


Figure 11: Log marginal likelihood vs. length scale. The optimum at $\ell \approx 0.95$ represents the best bias–variance tradeoff. This enables hyperparameter selection without cross-validation.

5.2 Effect of Noise Level σ_n

Figure 12 shows the effect of σ_n on the regression fit. Very small σ_n (0.01) forces the GP to interpolate through every noisy observation, producing an overfit posterior with near-zero confidence bands at training points. Larger σ_n (1.0) increases the confidence band width and smooths the posterior mean, acknowledging that observations are unreliable. The true noise level ($\sigma_n = 0.3$) yields the best balance.

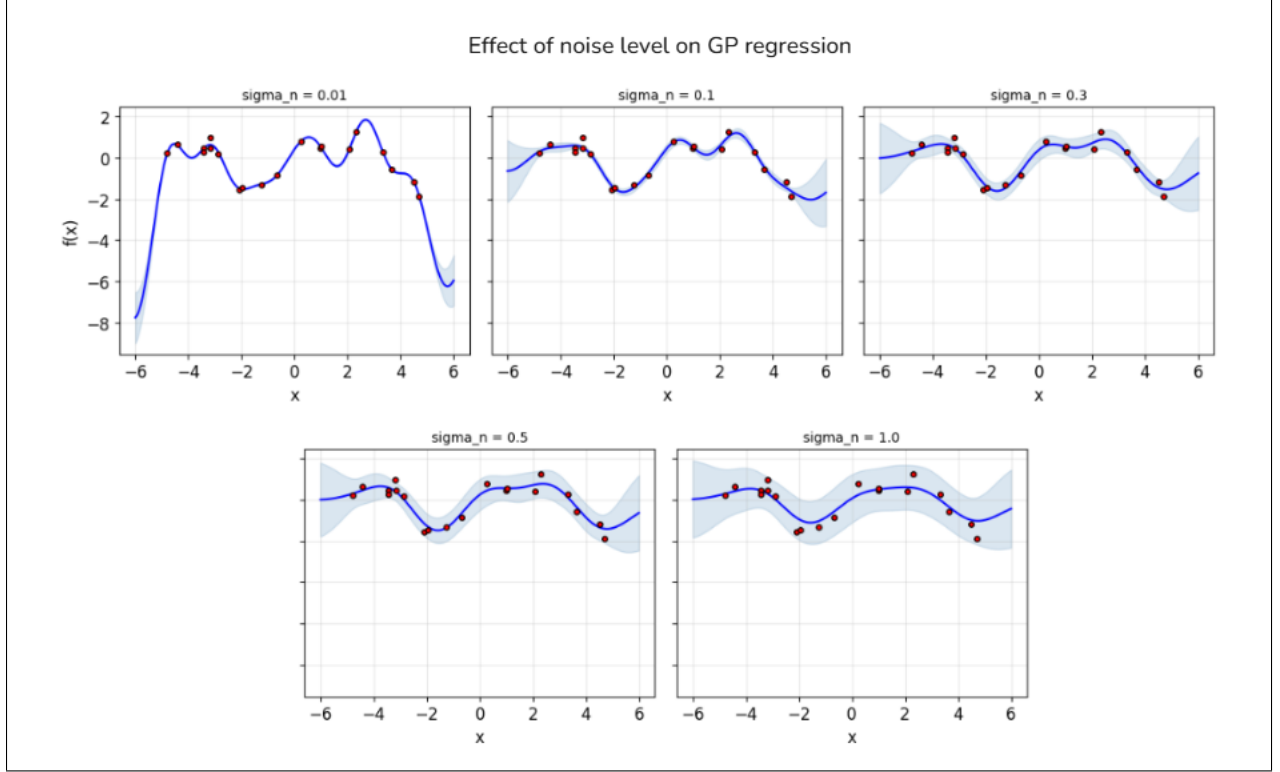


Figure 12: Effect of noise level σ_n on GP regression. Small σ_n causes interpolation (overfitting); large σ_n oversmooths.

5.3 Effect of Length Scale on GP Classification

Figure 13 shows how ℓ affects the GP classifier on Two Moons. Small ℓ (0.3) produces a fragmented, overfit decision boundary. As ℓ increases, the boundary smooths out, but at $\ell = 3.0$ it becomes nearly linear and fails to capture the moons' curvature. The latent variance maps (bottom row) show that variance is uniformly high for very small ℓ (since each point only influences a tiny neighbourhood) and uniformly low for large ℓ (since the GP is overconfident in its smooth fit).

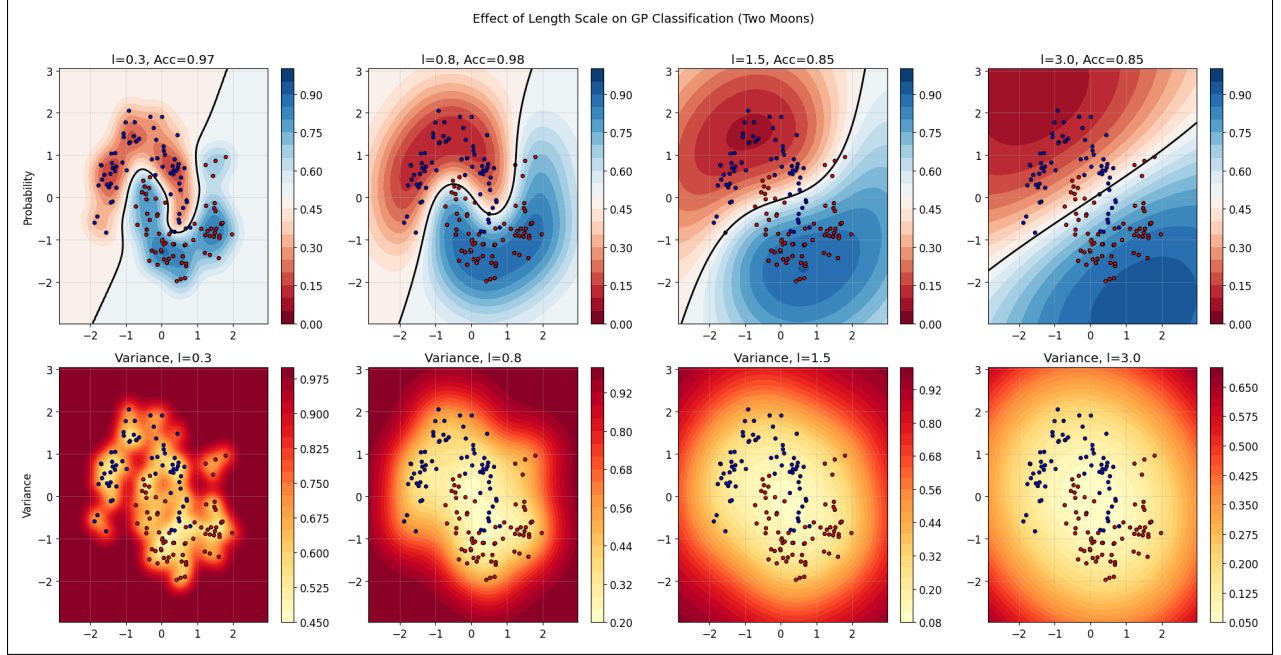


Figure 13: Effect of length scale on GP classification (Two Moons). Top: predictive probability. Bottom: latent variance. Optimal $\ell \approx 0.8$ captures the nonlinear boundary without overfitting.

5.4 Kernel Comparison

Figure 14 compares the RBF and Matérn 3/2 kernels at two length scales. The RBF kernel produces infinitely smooth functions, while Matérn 3/2 allows for slightly rougher functions. At $\ell = 1.0$, both kernels give similar fits, with the RBF achieving a marginally higher LML. At $\ell = 0.5$, both kernels track the data more closely, with the Matérn producing a slightly less smooth posterior.

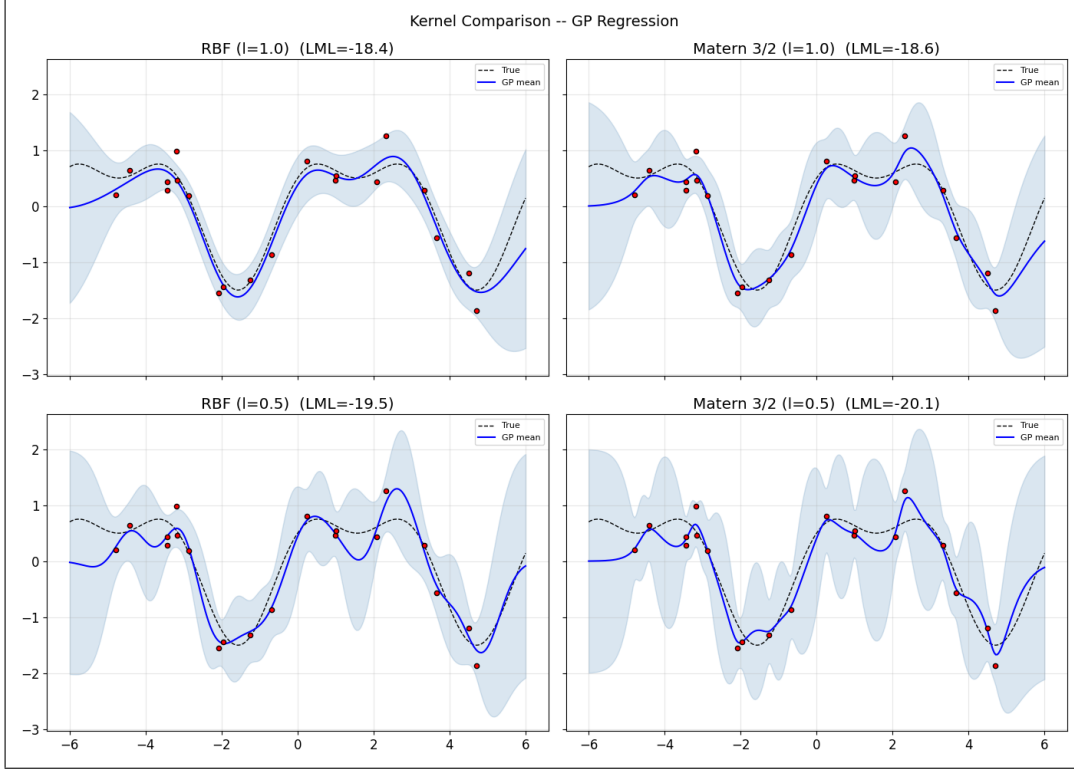


Figure 14: Kernel comparison for GP regression: RBF vs. Matérn 3/2 at two length scales. Log marginal likelihood values are shown in each title.

6 Additional Analysis

6.1 Confusion Matrices

Figure 15 shows confusion matrices for both classification experiments. On Two Moons (98.3% accuracy), only 1 test sample is misclassified. On Iris binary (97.8%), 1 sample is misclassified, indicating near-perfect separation of Setosa from the other two species using only two features.

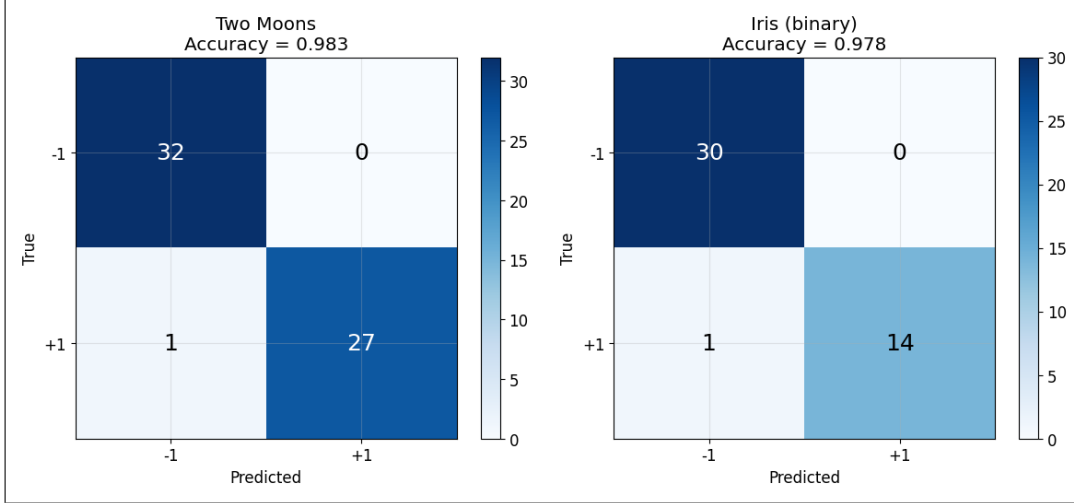


Figure 15: Confusion matrices. Left: Two Moons (98.3%). Right: Iris binary (97.8%).

6.2 Reliability Diagrams

Figure 16 shows reliability diagrams (calibration curves) for both datasets. A perfectly calibrated model would lie on the red diagonal. The GP classifier’s predictions track the diagonal reasonably well, indicating that the averaged prediction $\bar{\pi}_*$ provides well-calibrated probability estimates.

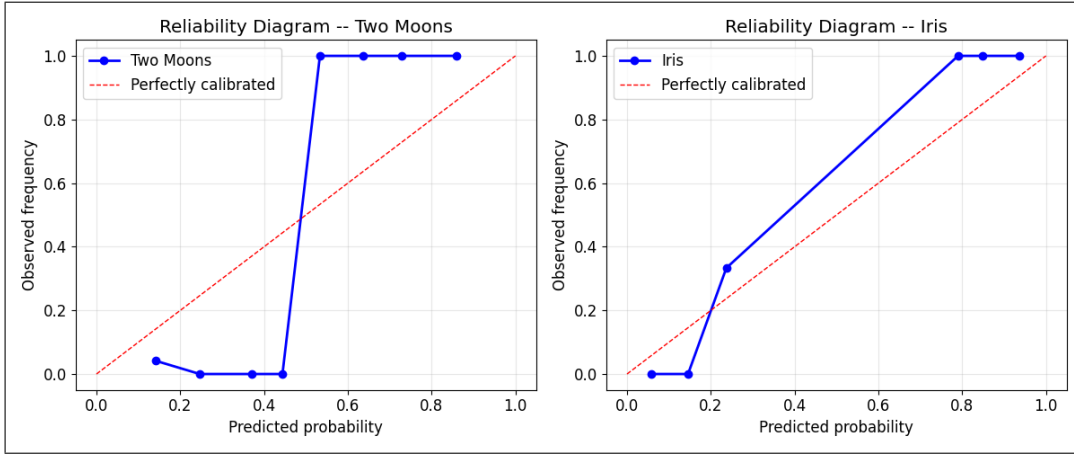


Figure 16: Reliability diagrams. Points near the diagonal indicate well-calibrated probabilities. Left: Two Moons. Right: Iris binary.

7 Discussion

- **Uncertainty structure.** The predictive variance $\mathbb{V}[f_*]$ in GP regression depends only on the test point locations and the kernel, not on the observed target values $\mathbf{y}$. This is a fundamental property of Gaussian conditioning. In classification, the latent variance additionally depends on the data through the Hessian W , since the Laplace approximation covariance is $(K^{-1} + W)^{-1}$.
- **MAP vs. averaged predictions.** The MAP prediction $\hat{\pi}_* = \sigma(\bar{f}_*)$ can be overconfident in

regions of high latent variance, because it ignores the spread of the posterior over f_* . The averaged prediction $\bar{\pi}_*$ integrates over this uncertainty, pulling probabilities toward 0.5 where the model is unsure. This effect is clearly visible in the difference plot (Figure 9, right panel).

- **Length scale controls the bias–variance tradeoff.** Small ℓ allows the GP to fit arbitrary local patterns (low bias, high variance), while large ℓ enforces smoothness (high bias, low variance). The log marginal likelihood provides an automatic Occam’s razor: it penalises both overly complex models (large $-\frac{1}{2} \log |K + \sigma_n^2 I|$) and poor data fit (large $-\frac{1}{2} \mathbf{y}^\top (K + \sigma_n^2 I)^{-1} \mathbf{y}$).
- **Newton convergence.** Algorithm 3.1 typically converges in 5–7 iterations on our datasets. This rapid convergence is expected because the log posterior is concave (it is the sum of concave log-likelihood terms and a concave log-prior), so the Newton method converges quadratically near the optimum.
- **Scalability.** The $O(n^3)$ Cholesky decomposition limits GPs to moderate dataset sizes. We used at most 300 training points for regression and 140 for classification. For larger datasets, sparse approximations (inducing points) or stochastic variational methods would be needed.

8 Implementation Notes

- All three algorithms are implemented from scratch. The only external dependencies are NumPy, SciPy (for `cholesky`, `solve_triangular`, `expit`), and scikit-learn (for datasets and metrics only).
- Two kernel functions are implemented: Squared Exponential (RBF) and Matérn 3/2, each parameterised by length scale ℓ and signal variance σ_f^2 .
- The probit approximation $\bar{\pi}_* \approx \sigma(\bar{f}_* / \sqrt{1 + \pi \mathbb{V}[f_*] / 8})$ is used for the Gaussian-sigmoid integral in Algorithm 3.2.
- Labels for classification are encoded as $\{-1, +1\}$ to match the logistic likelihood convention in Rasmussen & Williams.

9 Conclusion

We implemented Algorithms 2.1, 3.1, and 3.2 from Rasmussen & Williams and validated them on five datasets. The key findings are:

1. GP regression achieves $R^2 = 0.92$ on Friedman #1 (5-D) with well-calibrated uncertainty estimates, using only 300 training points and a single closed-form inference step.
2. The Laplace approximation for GP classification converges in 5–7 Newton iterations and achieves 98.33% accuracy on Two Moons, with the averaged prediction $\bar{\pi}_*$ providing more conservative and better-calibrated probabilities than the MAP prediction.
3. The predictive covariance visualisation in 2-D reveals how GP uncertainty increases with distance from training data and how nearby test points are positively correlated under the RBF kernel.
4. The log marginal likelihood serves as an effective model selection criterion, identifying the optimal length scale without cross-validation.

5. Hyperparameter ablations confirm that small ℓ overfits, large ℓ undersmooths, and the kernel choice (RBF vs. Matérn) has a secondary effect compared to proper length scale selection.